JEA/SC
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Installation of the press

It is essential that the foundation on which the press and its motor are based, is of a heavy and rigid construction and is made according to the dimension drawing from Stord International.

To facilitate the transport, the press can be partly dismantled before dispatch from the workshop, where it has been run in and checked before being dismantled and packed for shipment.

When fitting up more than one press simultaneously, one should take care not to mix components from presses of different production numbers.

To facilitate handling and lifting of the separate parts, a trevelling hoist or tackle must be installed above the press.

Recommended lifting capacity is minimum 7 tons.

Check that all machined surfaces are clean and without damages.

Mounting and fitting of the press on site are carried out as follows:

1. Check that the premade foundation is in level.

   Take off the covers and the upper cages.

   Place the press on the premade foundation. If this foundation is made of concrete with holes for the bolts, suitable soft cement must be poured into the holes under the baseplate. The assembly can be continued only after a drying period of min. 3 days.

2. The press should be accurately levelled by means of optic levelling instrument, please see drawing no. 21–41829.

   The measuring point for new presses is the press frame, and for secondhand presses the side bars.

   Level one side first, and then the opposite side by using the adjustment screws on gearbox and supports. When the opposite side is correct, please check the starting side for any adjustments.

   Three measuring points on each side of the press is satisfactory.

3. Steel spacers should be put under the gearbox and support. The steel spacers have to be accurate and cover the area of the bedplate.

   The foundation bolts to be tightened.

   Please check the levelling again and make adjustments if necessary.

The following sections 4 and 5 should be followed when the press screws are dismantled or sent separately.

4. Place the press screw in the cage as indicated on the drawing, and take care that the screw flights mesh correctly, i.e. each flight of the one screw must be in position midway between two flights of the other screw.

   Each press screw can now be pushed onto the shaft journal until it lies true against the flange, whereupon the screw bolts are fastened thoroughly. Make sure that the abutting flange faces are absolutely clean and that the markings on the screw body correspond with those on the flange.

   It is very important that the flange screws M36 are torqued to 2.200 Nm (1.600 ft. lbf.). Should it not be possible to check the degree of adjustment, the angle of incidence ("turn of nut") can be measured from the fixed installation of the screws ("snug fit"). For this particular dimension of screws, ½–¾ turns. The screws are secured by lock plates.

5. The press screws are lifted at the inlet end, and the lower part of the bearing housings is fitted onto the press by means of dowels and bolts. The thrust sleeve and the bearings are fitted onto the shafts, and each screw is lowered into place.

   Fit the bearings in the middle of the housing and fix by means of the adapter sleeves. Fit the upper part of the housing and fill with grease, then fit the inlet chute . The end plate is fixed by means of conical dowels. Then fit the seal rings and seal house onto each shaft journal.

6. Air blast the lower part of the press cage to remove dirt and foreign matters.

7. Turn the press screws and check by means of a feeler the clearance between the flights and the strainer plates in the lower part of the cage all over the length of the screws. The clearance should be minimum 0,8 mm (0,03").

Pay special attention to the clearances in the lower cage in the mid section. It may be necessary to lower the press on the middle to obtain satisfactory clearance here. Foundation bolts on the middle legs can now be tightened.

8. Position the 2 upper parts of the press cage. Insert the distance bars and fasten them. Clamp the upper part of the press cage to the lower part by means of the sling clamps. These should be pulled tight.

9. To facilitate future dismantling of inlet chute in press, there should be arranged for ample free height (at least 470 mm) above the flanged inlet on the press, when the outside chute is removed.

10. Turn the gearbox input shaft by hand. This should turn easy and even. The clearance between screws and upper cage is checked visually with a hand lamp by looking through the perforated plate and turning the press slowly by hand. Listen for sounds indicating the screw touching the strainer plate. The clearance can also be checked by means of a special depth indicator inserted through the perforations.

11. Make sure that all bolts and nuts are tightened.

12. Fit the driving motor and motor for oil pump. The electrical starters for hydraulic unit for press and oil lubrication pump must be connected in a way that the hydraulic unit/driving motor should only be able to start after the oil lubrication pump is running. The driving motor should automatically be switched off if the oil lubrication pump motor cuts out.

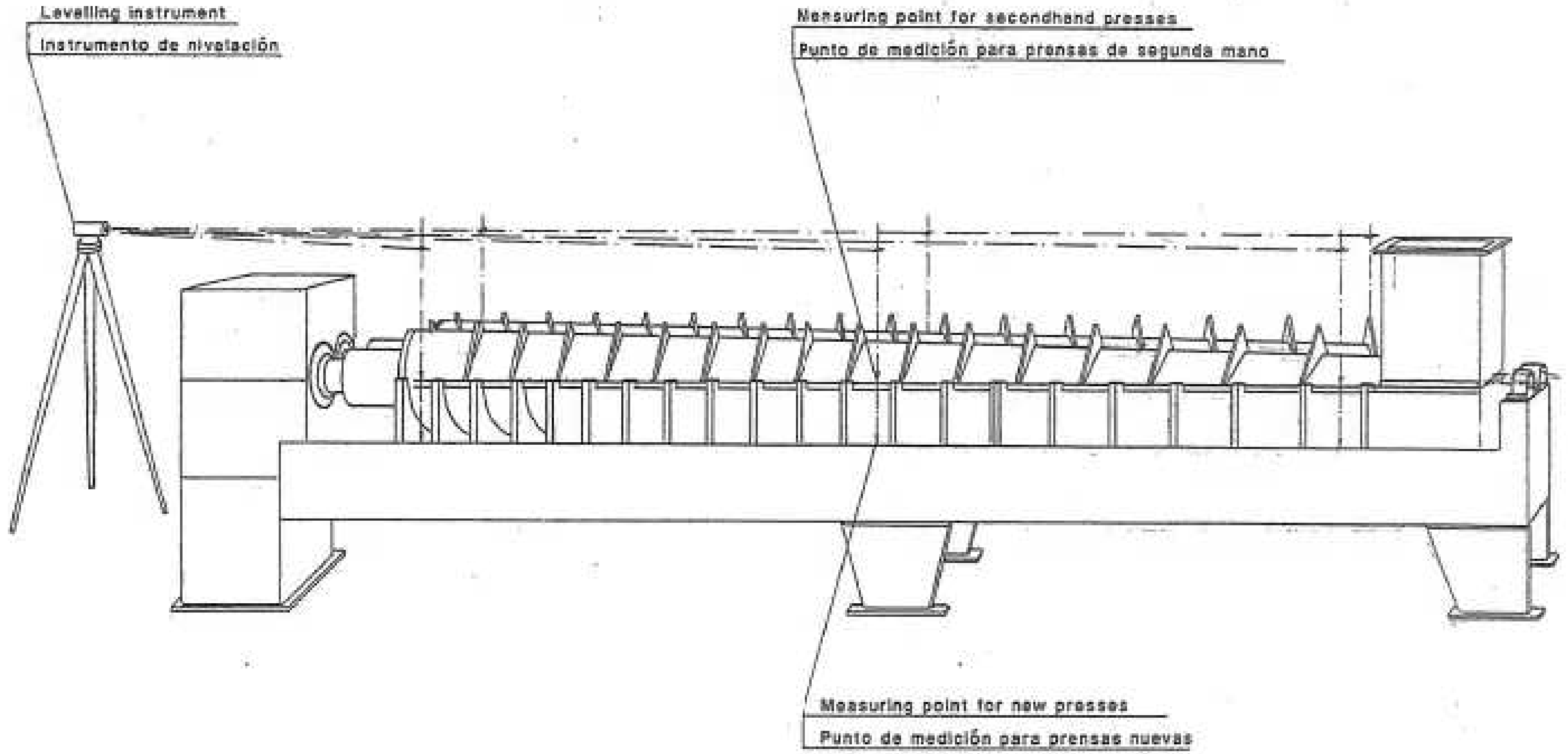
13. Fill oil according to lubrication instructions. For the hydraulic drive unit, please see separate instructions.

    NB! Please make sure that oil pipes for hydraulic drive are completely cleaned.

14. Mount the covers ofer the press cage. For presses equipped with flushing pipes, these should be fitted before the covers are positioned.

15. Start the press for trial running by engaging the motor, and cut out immediately if anything appears to be wrong. Check and start again.

    When starting press in ambient temperatures below 10 °C (50 °F), it is recommended to by–pass the filter for about 1–2 hours, in order to prevent damage to the cartridge by the thick cold oil.



Press RS/MS-64
Alignment
Alineación

Stord International a.s
BERGEN - NORWAY

Nr.
No. 21-41829

Lubrication

Lubrication of Gearbox:

The gearbox has been thoroughly cleaned before dispatch from the works and is ready for filling up with the required quality and quantity of oil up to the middle of the oil level indicator.

The gearbox, bearings, gearwheels and lubrication system have been treated with a rust–inhibiting oil which does not require cleaning off before filling the oil. The recommended quantity and quality is to be found on a separate Lubrication Chart further back in this instruction manual. The gearbox has its own lubricant system. The oil pump sucks from the oil pan in the gearbox and pumps it through the strainer to an oil distributing tank in the upper part of the gearbox.

From the oil tank the oil is distributed over the gear wheels and to the bearings and flows back to the oil pan.

Regular checks on the oil pressure from the pump and flow of oil from the two large thrust bearings are recommended. (Visual inspection through the transparent side covers).

For filtration of the oil, the system is equipped with filters. One coarse (inside the gearbox) on the suction side of the pump and a finer one on the delivery side. The finer filter has a valve allowing the oil to flow in by–pass during cleaning of the filter and bowl.

During running–in of the press it is recommended to clean the external filter after the first 30 hours. Later every 200 hours, or when the gauge shows pressure in excess of 2,8 bar (40 psi).

(When starting up with cold oil the pressure will initially exceed this value). Cleaning is done by loosening down the bowl and cleaning the brass gauze cartridge in petrol or paraffin. For cleaning use a soft brush. Wipe off the magnet with a clean rag, check that the filter cartridge is in working order, replace the filter and reset valve for filtration. Spare cartridges should always be in store.

The oil in the gearbox should be changed for the first time after 2000 working hours, later after every 5000 hours or according to sample analysis by the oil company. The internal suction strainer should be cleaned at every oil change.

If the oil temperature exceeds 60 °C (140 °F), it will be necessary to install an oil–cooler. Particulars from Stord International A/S.

The oil pump motor is to be lubricated according to normal practice.

Lubrication Control and Protection:

If the press is to be run without regular supervision, it is recommended to fit a pressure switch with a warning and motor cut out system.

The external oil pipe is in front of the oil filter equipped with a 3/4" int. pipe thread junction for the possible fitting of a pressure switch. On delivery, a 3/4" blanking plug is screwed in. The warning system should give alarm at oil pressure below 0,25 bar (3,5 psi) and the interloc. should automatically shut down the driving motor.

Pressures above 2,8 bar (40 psi) indicates that the oil filter is due for cleaning.

(When starting up with cold oil, the pressure will initially exceed the above mentioned value.)

Starting up at low Temperatures:

At low ambient temperatures, 5 °C (40 °F) or lower, the oil can be so viscous that the pump will have difficulties in pumping the lubricant and in this case the oil should be heated some–what, primarily at the pump intake. This can be done for instance by using a large fan heater to blow hot air against the gearbox at the pump intake. If the press is erected outside, a preheater or an electrical spiralheater should be installed.

Press Screw Bearings Inlet End:

The two bearings at the inlet end are lubricated by grease. Fill the housing completely with grease when fitting this and refill once every other week with a grease gun. The bearings should be cleaned and replenished with grease annually.

Oilseals on Output Shafts from Gearbox:

It is recommended to grease these seals once a year.

This is done by unscrewing both of the plugs in the cover (item 14) and fitting a standard grease nipple in one of the holes.
Press grease until it flows out of relief hole. Now refit the plugs.

Operation

When the erection of the machine is completed and oil filled, start the press and let it run for about 1 hour before charging. When starting press in ambient temperatures below 10 °C it is recommended to by–pass the filter for about 1–2 hours. This to prevent damage to the cartridge by the thick cold oil. Start feeding with the motor running at lowest speed, and be sure that there is a suitable level of raw material in the inlet chute. Generally normal pressing conditions are not obtained before the press is completely filled and the initial charge has gone through. If feeding is insufficient, the pressure will decrease and the water content in the discharge will increase. The best results are obtained when the press is working at a constant speed with a continuous and uniform supply.

Watch the kW or Amp. meter and make sure that the electrical loading is not higher than the electric motor rate.

To avoid risk of damaging the machine as a result of clogging and encrusting, the press should be emptied when taken out of operation. However, if the press is stopped for emergency not exceeding half an hour, the material may be left in the press.

If the press stops due to overload/jamming, lift up the upper part of the press cage (at the outlet end), allowing some of the material to be removed before the press is put into operation again.

IMPORTANT!

Do not in any case try to run the press screws in reverse, as severe damage of the gearbox then can occur.

Generally, stones and small pieces of iron will do no harm to the press. Bigger pieces, however, might cause damage to the strainer plates. In case of damage, stop the press at once, lift off the top part of the press cage, and remove the material. Replace strainer plates as described in special instruction. In case of minor damage to the strainer screen the press can be run empty, and by provisionally blinding off the damaged area from the outside, the press can quickly be put in operation again.

Inspection and maintenance

1. During Operation:

From time to time the press should be inspected and fasteners tightened if necessary. Deposits of material on the press cage, frame or other places should be hosed off.

The oil pressure and temperature should be checked regularly. The filter should be cleaned for the first time after 30 operating hours. Later – under normal operating conditions – every 200 hrs. Should the oil pressure exceed 2,8 bar (40 psi) the filter should in any case be cleaned or changed.

The oil in the gearbox should be changed for the first time after, say, 2000 operation hrs. Later every 5000 hrs or according to analysis.

2. After Production:

For extended shut–down periods, the press should be hosed with water [pressure up to 50 bar (750 psi)] in order to avoid material remainder fermenting and causing corrosion. Further cleaning and maintenance should take place as soon as possible.

Cover guard sections and shutters should be properly cleaned and brushed on the inside. Hot water should be used. Check that the seals are in order.

Once a year the press cage should be opened for examination of the strainer plates.

A suitable, spacious place in the works should be reserved for the placing of parts during overhaul work. Special attention should be paid to machined parts, which should be placed on wood supports.

Cleaning of the Press Cage:

The upper part of the cage sections should be detached and cleaned without further dismantling unless damaged strainer plates have to be replaced or if there should be signs indicating corrosion on the contact surface between the backing plates and the bridges. Hot water should be used for cleaning, preferably at high pressure. Material remainder sticking to the perforation can be removed by means of a steel brush and a scraper and blown out by pressurized air or water.

The easiest way to clean the lower part of the cage is to remove the backing plates. See section: Removal and Remounting of Backing plates.

Frames with strainer plates under the inlet chute can be cleaned in position in the press. However, these frames can easily be removed after the inlet chute has been dismantled. Before assembly, it is important that the surfaces of the bridges mating the bars and grooves in the side bars are thoroughly cleaned, and care must be taken to ensure that the plates are mounted in the correct place.

Strainer Plates:

Minor damage to the strainer plates can often be made good by welding (e.g. Argonarc method). Afterwards the welding is ground down carefully.

Painting:

Good maintenance of the press is worthwhile, and too much time should not elapse before it is repainted. It is important that grease, dirt and rust are removed first. It is advisable to use a good primer. Information onf suitable qualities of paint can be supplied upon request.

Press Screw Flights:

The top of the press screw flights are on some materials subject to wear and the leading edge should be rebuilt by welding when so worn that the pressing effect is influenced. This depends somewhat on the nature and condition of the material to be pressed.

Before welding takes place, a suitable backing strip should be clamped to the screw flight, as shown on fig. 41-33430. This strip may be of copper (not annealed) or carbon. The latter is preferable, because the weld deposit sometimes has a tendency to stick to the copper backing. This may, however, be avoided to some degree if the arc is not directed at the copper. As regards the choice of welding electrodes, we are always at your service.

Lubrication of Gearbox:

The gearbox should be flushed thoroughly when the oil is changed. Clean cloths of cotton or linen can be used for drying. Cotton waste (twist) must on no account be used.

The type of oil is specified in the lubrication chart.

Protection of the Gearbox in Lay–off Periods:

Due to condensation and possible acid formation internally in the gearbox during lay–off periods, it is necessary to protect roller bearings, gearwheels and lubrication system against corrosion. Preparing for a long lay–off period, the oil should therefore be drained immediately after the press has been taken out of commission, and the gearbox and filters should be cleaned. The gearbox is then filled with rust–inhibiting oil (for instance Mobilarma 524) to the normal oil level. The press is then run idle for 10–15 minutes with this oil, the oil is then drained and can be used in the next press or taken care of till next time. It is not necessary to clean off this oil before normal lubricant is filled, before putting the press into commission again.

Alternatively, the gearbox can be protected against rusting by leaving the lubrication oil in and then once every two weeks, starting the press and idling it for 5–10 minutes. Before starting, drain out possible water that could have gathered in the gearbox.